

Improved Training Sample Efficiency and Inter-Device Generalizability in Optical Coherence Tomography Fluid Segmentation via Foundation Models

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Abstract. Accurate segmentation of fluid regions in optical coherence tomography (OCT) images is crucial for ophthalmologic diagnosis and treatment monitoring. However, automated segmentation models face two key challenges: high annotation costs and limited generalization across OCT devices. We investigate foundation models to address these challenges, evaluating a domain-specific OCT foundation model trained using the SimCLR method and an adapted Segment Anything Model 2 (SAM2). Through experiments with datasets ranging from 50 to 2,500 images and cross-device validation, we show that foundation models outperform ImageNet-pretrained models in small data regimes. In cross-device evaluation, both foundation models demonstrated superior generalization. Our findings indicate that foundation models significantly reduce annotation requirements and enhance cross-device adaptability, lowering development costs and accelerating deployment of OCT fluid segmentation solutions.

Keywords: Optical Coherence Tomography · Foundation Models · Medical Image Segmentation · Sample Efficiency · Generalizability.

1 Introduction

In the treatment and management of retinal exudative diseases such as neovascular age-related macular degeneration (nAMD) and diabetic macular edema (DME), the accurate measurement of fluid in the retina is crucial for making treatment decisions and predicting the treatment outcome [12, 4]. Optical Coherence Tomography (OCT) is a non-invasive imaging modality, operating at micrometer-scale resolution, which enables the detailed visualization of retinal

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structure and pathology, particularly the accumulation of fluid in different retinal layers.

The process of manually annotating fluid regions in OCT scans is labor-intensive and time-consuming. Expert graders, typically experienced ophthalmologists or trained technicians, must meticulously delineate fluid boundaries across numerous B-scan slices. This labor-intensive process is prohibitive in real world clinical settings [2]. The challenge necessitates the development of automated systems that can segment the fluid regions in OCT scans. Previous works have shown promising results using deep learning [14]. However, collecting the manual annotations is still a significant bottleneck in the development pipeline.

Additionally, the field faces a challenge in achieving reliable inter-device generalizability, as OCT imaging characteristics can vary significantly across different manufacturer platforms [5]. For instance, Spectralis (Heidelberg Engineering, Inc.) and Cirrus (Carl Zeiss Meditec, Inc.) systems produce images with distinct resolution profiles and noise characteristics (Figure 1). Mitigating the degradation and improving model generalizability is highly desirable because device-specific training requires a costly process of collecting the manual annotations for each device.

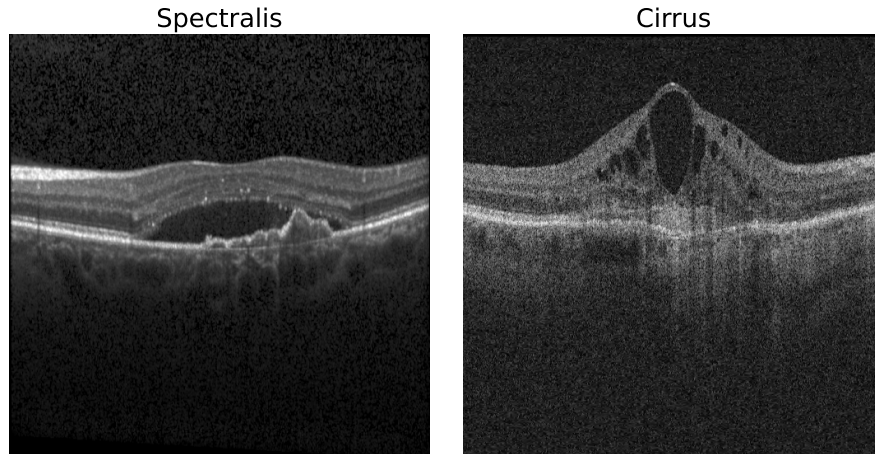


Fig. 1. Sample images from the Spectralis (Heidelberg Engineering) and the Cirrus (Carl Zeiss Meditec) OCT devices. Both images were resized to 512 by 512 pixels from the original size.

A foundation model is a model pretrained on a large heterogeneous dataset typically using self-supervised learning methods to acquire the general-purpose representation. The large scale pretraining of foundation models often eases the requirements on the size of the task-specific training set and exhibits better generalizability to unseen data [1]. In this study, we evaluate two distinct approaches of foundation model-based OCT fluid segmentation on the challenges

in the training size requirement (sample efficiency) and inter-device generalization. The first approach utilizes an internally developed foundation model trained using the SimCLR method [3], specifically trained on a large corpus of OCT B-scans to capture domain-specific features. The second approach leverages Segment Anything Model 2 [11] architecture, a publicly available foundation model originally designed for general-purpose image segmentation.

While most foundation model research focuses primarily on absolute performance metrics, sample efficiency and out-of-distribution generalization are two critical aspects that have been less frequently and thoroughly addressed. Our analysis is centered around these two aspects in the task of OCT fluid segmentation. Specifically, we address the following questions through carefully designed experiments.

Q1. (Sample efficiency) How much labeled data for training segmentation models can be saved by leveraging foundation models compared to the standard approach without foundation models?

Q2. (Inter-device generalizability) What is the extent of out-of-distribution generalizability of foundation models compared to the generalizability of the standard approach without foundation models? Additionally, how does the performance of foundation models compare to the performance of the in-domain standard approach?

2 Methods

2.1 OCT Fluid Segmentation

Our investigation centers on the challenging task of pixel-level multi-class fluid segmentation in OCT images, specifically focusing on three distinct fluid types: Intraretinal fluid (IRF), subretinal fluid (SRF), and pigment epithelial detachment (PED). To conduct the sample efficiency and inter-device generalizability experiments, we used two different datasets.

- D_S : Spectralis dataset. The dataset consists of OCT B-scans acquired using Spectralis (Heidelberg Engineering) devices in AVENUE clinical trial in nAMD (NCT02484690) and BOULEVARD clinical trial in DME (NCT02699450). The dataset was split into training, validation, and holdout testing. There are 2,568 B-scans for training, 458 B-scans for validation, and 371 B-scans for holdout testing.
- D_C : Cirrus dataset. The dataset consists of OCT B-scans acquired using Zeiss (Carl Zeiss Meditec) devices in HARBOR clinical trial in nAMD (NCT06411288). HARBOR is a clinical trial in nAMD. The dataset was also split into training, validation, and holdout testing. There are 735 B-scans for training, 126 B-scans for validation, and 99 B-scans for holdout testing.

All the splits were done at the patient level to avoid any potential data leakage. The annotation was done by trained graders and the quality was reviewed by a senior grader. More detailed protocol is available in [10]. The roles of the datasets are described in 2.4.

2.2 In-domain Foundation Model Development

SimCLR Foundation Model Following the SimCLR framework [3], our model was trained on a large dataset of unlabeled OCT B-scans, which include clinical studies in nAMD, DME, and geographic atrophy (GA). The OCT scans were taken by devices manufactured by Heidelberg Engineering, Inc., Carl Zeiss Meditec, Inc., and Topcon Corporation, Inc. The training data were curated to exclude any patients whose data are in the validation and the holdout sets of D_S and D_C .

The training set consists of 765,586 OCT volumes from 9,509 patients (27% nAMD, 37% DME, and 36% GA). The device distribution is 70% Spectralis, 26% Cirrus, and 4% Topcon. To prepare the training data for SimCLR pretraining, individual B-scans were extracted from OCT volumes. Because the B-scans taken at off-central locations are less likely to contain pathological biomarkers and the appearance is similar, only the B-scans from the middle 50% on the transversal direction were included. As a result, the training set consists of 18,916,869 B-scans.

A ResNet50 [6] model was used as the backbone model. The SimCLR training followed the original framework including the input image size of 224 by 224 pixels. The batch size was 6,000 and the number of epochs was 40. The weights of ResNet50 were initialized with ImageNet weights.

Adapting Foundation Model for Segmentation The SimCLR model only consists of the encoder part and therefore we need to convert it to a segmentation model by adding a decoder part. Based on the popularity and the proven performance of the U-Net architecture [13] in medical image segmentation, we chose to convert the ResNet50 model to a ResUNet [7] architecture. In fine-tuning on the segmentation task, the decoder part was randomly initialized and the entire model was trained end to end.

Regarding the training of ResUNet, we made two observations: 1) Some target masks, typically IRF, could be very small in the size of 224 by 224 pixels. Concretely, the majority of the IRF masks were less than 0.2% area of the image, which is approximately 100 pixels in the image size of 224 by 224 pixels. 2) At the beginning of the training, the weights of the encoder are already optimized after the SimCLR pretraining while the decoder part is freshly initialized. To address 1), we used a larger image size of 512 by 512 pixels for fine-tuning. For 2), a different learning rate for the encoder and the decoder was included in the hyper parameter search (the options of learning rate are 1e-4 and 1e-5 for the encoder and 1e-3 and 1e-4 for the decoder). Due to the limitation in time and computational resources, we used the default parameters for the following parameters: equally weight Dice and focal loss for the loss function, AdamW [8] for the optimizer, linear warm up and cosine annealing [9] for the learning rate scheduler, and the batch size of 20.

2.3 SAM2-Based Approach

Segment Anything Model 2 (SAM2) is a foundation model for general-purpose image segmentation. It was trained on a large dataset of annotated natural videos [11]. We adapted SAM2 for the OCT fluid segmentation task by closely following the SAM2-UNet architecture and implementation [15]. The SAM2-UNet architecture is comprised of the SAM2 encoder and a convolutional decoder where the intermediate feature representation from different resolutions are connected to the decoder by skip connections, which makes the whole architecture a U-Net [13] shape. To facilitate efficient training, the authors of SAM2-UNet employed a parameter-efficient fine-tuning approach and structure loss that combines intersection-over-union loss and cross-entropy loss that highlights the boundaries [15]. The two observations about the image size and the learning rate for the encoder and decoder were also examined for SAM2-UNet as well. For the image size, the default image size for SAM2 of 1024 by 1024 pixels was used because our concern was about small image size. For the learning rate, the trainable parameters in SAM2-UNet are the adaptors and the decoder. Because all of them are randomly initialized, we used the same learning rate for the encoder and the decoder. We tuned the learning rate (1e-3 and 1e-4) and the kernel size in the structure loss (31, 41, and 51). Again, due to the computational resource limitation, we fixed the following parameters: AdamW [8] for the optimizer, cosine annealing [9] for the learning rate scheduler, and the batch size of 4.

The model structure of adapting the foundation models and the ImageNet pretrained model is shown in Figure 2.

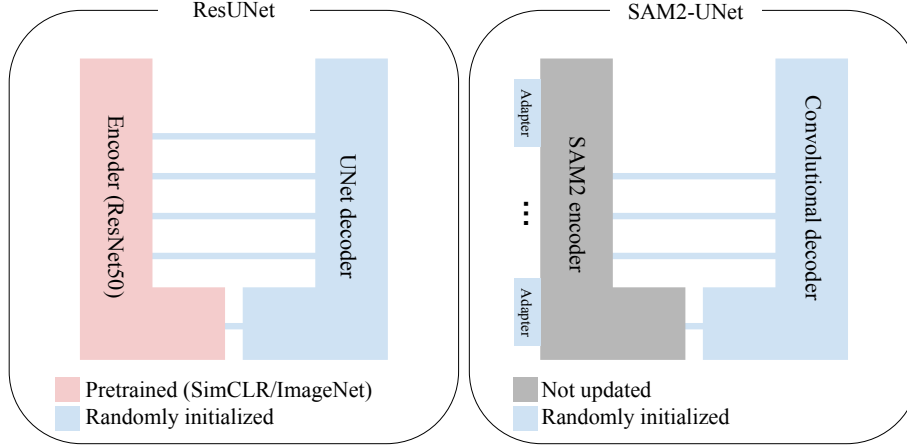


Fig. 2. Model structure of adapting the foundation models and the ImageNet pretrained model. ResUNet (left) and SAM2-UNet (right).

2.4 Experimental Design

Sample Efficiency Evaluation The sample efficiency experiments were designed to evaluate model performance across different training data regimes. The following five distinct dataset proportions of the training set of D_S were used: 100%, 40% (= 1,028 $\approx$ 1,000 B-scans), 20% (=514 $\approx$ 500 B-scans), 4% (=103 $\approx$ 100 B-scans), and 2% (= 52 $\approx$ 50 B-scans). The validation and the holdout sets were kept the same regardless of the training set size. To ensure the training set contains representative cases, we stratified the training set by the total fluid volume and then randomly sampled the cases from each stratum for five distinct data portions. In hyper parameter tuning, we ran the training process five times with different random seeds to average out the randomness. The models with the hyper parameter combination with the lowest validation loss averaged over the random seeds was selected to be evaluated on the holdout test set.

Inter-device Generalizability Assessment The evaluation of inter-device generalizability focused on zero-shot performance of models trained on images taken on Spectralis devices when applied to images taken with Cirrus devices. The models trained on 100% of D_S training set that were selected for the holdout testing were evaluated on the holdout testing set of D_C .

2.5 Evaluation Metrics and Baseline Models

The final model performance on the holdout set was evaluated using classwise Dice score excluding the background class. Note that the Dice score when both the target and the prediction are empty is defined to be 1.

To highlight the added values from foundation models, we compare the performance of the foundation models with the baseline models. We chose the ResUNet model with the encoder initialized with ResNet50 ImageNet weights as the baseline model. The training details are the same as the ResUNet models with SimCLR ResNet50 encoder. Hereafter, we refer to the ResUNet model with the initial weights pretrained on our internal data using the SimCLR as SimCLR ResUNet and the ResUNet model with the initial weights from ImageNet as ImageNet ResUNet. Also, we collectively call SimCLR ResUNet and SAM2-UNet as foundation model-based approaches.

We also added ImageNet ResUNet trained on D_C to the comparison of the inter-device generalizability experiments. This addition enables us to compare the performance of foundation model-based approaches with a standard approach in D_C when training data is available.

2.6 Implementation Details

All the experiments were conducted on NVIDIA A100 GPUs with 40GB of memory. The SimCLR ResNet50 was trained using 16 GPUs. The ResUNet models and SAM2-UNet models were trained using a single GPU.

3 Results

3.1 Sample Efficiency Analysis

Our evaluation of sample efficiency revealed several significant findings across different training data regimes. The results demonstrated a clear advantage of foundation model-based approaches over traditional training methods, particularly in low-data scenarios. We look at the results from the two comparisons: 1) ImageNet ResUNet vs. foundation model-based approaches and 2) SAM2-UNet vs. SimCLR ResUNet. For the first comparison, the foundation model-based approaches consistently outperformed the ImageNet ResUNet and showed more robust performance as the training data size decreased as shown in Figure 3. Examining the same figure, the foundation model-based approaches already reached the performance level at 100% with 40% training data and there are slight drops at 20% from 40%. On the other hand, the ImageNet ResUNet performance starts to drop at 40% compared to the performance at 100%. This would mean that the training size needed to fine-tune the foundation models is in between 500 and 1,000 samples and the ImageNet ResUNet needs more than 1,000 samples to fine-tune. For the second comparison within foundation model-based approaches, Figure 3 shows that the SAM2-UNet outperformed the SimCLR ResUNet with 4% to 100% of the training data in all fluid types while the relative rank changes with 2% of training data. The reason why SAM2-UNet exhibits a more acute performance drop than SimCLR ResUNet from 4% to 2% is not entirely clear without further experiments. However, a potential factor is the differences in the pretraining dataset. Because SAM2 was trained on natural images, it may need more training data to adapt to the OCT images than SimCLR ResUNet, which was trained on OCT images.

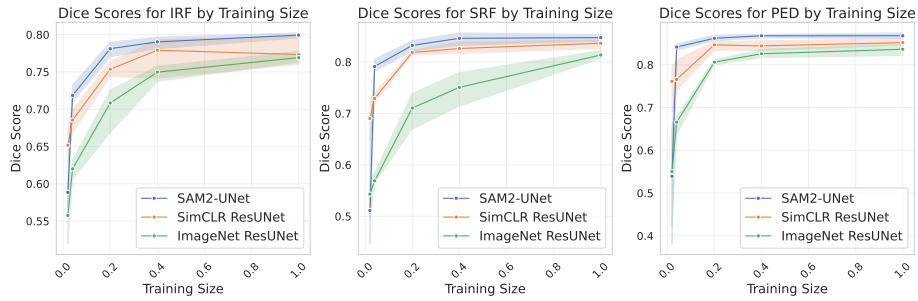


Fig. 3. Sample efficiency results. The mean holdout test Dice score for 5 random seeds is shown. The shaded area represents the highest and lowest mean Dice score out of the 5 random seeds.

3.2 Effect of Image Size and Learning Rate

Using a different image size in fine-tuning from pretraining is not standard practice. However, our results suggest that the fine-tuning process could help the models adapt to the larger image size.

Additionally, we report that a smaller learning rate was always preferred for the encoder part than the decoder part regardless of the initial encoder weights in the hyper parameter tuning of ResUNet.

3.3 Inter-device Generalizability

The models evaluated on the holdout testing set of D_S in the sample efficiency analysis were evaluated on the holdout testing set of D_C in a zero-shot way. The zero-shot performance was compared to the supervised ImageNet ResUNet model trained on D_C . The holdout test results are shown in Table 1. Overall, the foundation model-based approaches showed superior performance compared to the zero-shot and in-domain ImageNet ResUNet models. Among the foundation model-based approaches, SimCLR ResUNet and SAM2-UNet are similar in SRF and PED while SAM2-UNet showed strength in IRF. Looking at the predicted masks shown in Figure 4, the foundation model-based approaches respect the topology of the fluid regions, which implies those models learned it in the source domain in a generalizable way. The results indicate the strong generalization capability of the foundation models across different OCT devices.

Table 1. Inter-device generalizability results. The mean holdout test Dice score over 5 random seeds on the holdout testing set of D_C . The model name follows the form “{model architecture} on {fine-tuning dataset}”. The values in parentheses are the minimum and maximum Dice score out of the 5 random seeds.

Model	Dice score (IRF)	Dice score (SRF)	Dice score (PED)
SimCLR ResUNet on D_S	0.71 (0.68, 0.75)	0.71 (0.67, 0.73)	0.64 (0.60, 0.70)
SAM2-UNet on D_S	0.76 (0.73, 0.79)	0.72 (0.70, 0.74)	0.64 (0.60, 0.69)
ImageNet ResUNet on D_S	0.69 (0.66, 0.71)	0.45 (0.39, 0.57)	0.42 (0.34, 0.49)
ImageNet ResUNet on D_C	0.70 (0.65, 0.72)	0.55 (0.48, 0.67)	0.59 (0.57, 0.61)

4 Discussion and Conclusion

4.1 Key Findings and Implications

Our investigation into foundation model-based approaches for OCT fluid segmentation has revealed several significant findings. The superior performance of foundation model-based approaches suggests a potential for significant reduction in annotation requirements for future applications, which enables less time and cost for collecting data and ultimately leads to shorter development time.

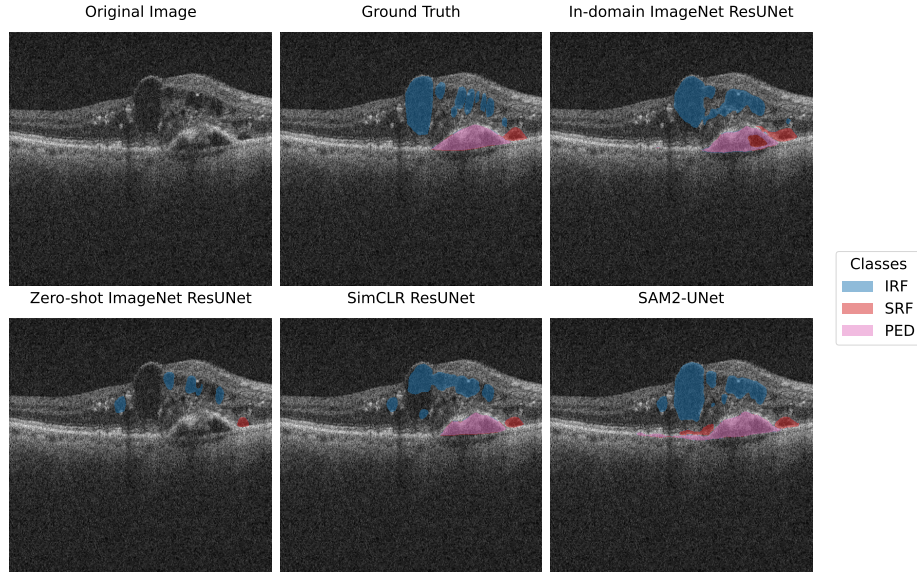


Fig. 4. A representative case highlighting the qualitative differences between different approaches. The in-domain ImageNet ResUNet predicted SRF inside PED, which violates the topology of the fluid regions. The Zero-shot ImageNet ResUNet missed a large portion of IRF and completely missed PED. The overall quality was improved in foundation model-based approaches although SimCLR ResUNet missed part of the IRF region and SAM2-UNet oversegmented.

In this study, we included two foundation models with different pretraining datasets and methods. SAM2 showed a pronounced domain adaptability in OCT fluid segmentation and outperformed SimCLR ResUNet in most of the cases except the very small data regime with about 50 B-scans for training. Even though the model size is a magnitude smaller, SimCLR ResUNet showed the advantage of in-domain pretraining by following SAM2-UNet closely in most of the experimental settings.

It is also notable that the SimCLR ResUNet consistently outperformed the ImageNet ResUNet in all experimental settings, which solely came from the difference in the initial weights. This supports that in-domain pretraining is advantageous in the context of medical image analysis, which aligns with the findings in the previous works such as [16].

4.2 Limitations and Future Works

In our study, we evaluated the two foundation models in the context of OCT fluid segmentation with a selected adaptation strategy. To validate and generalize our findings more broadly, it is warranted to evaluate more foundation models and adaptation strategies on a diverse range of tasks.

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